Paleomagnetism of Upper Triassic Diabase from Southeastern Pennsylvania: Further Results

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Seventy-eight reliable sites in Upper Triassic diabase from southeastern Pennsylvania yield a paleomagnetic pole at 62.0°N, 104.5°E, about 0.5° from an earlier pole calculation based on a preliminary study of 20 sites. Site poles have a Fisherian distribution, with an angular standard deviation of 7.4°, considerably less than the dispersion predicted by models based on the present geomagnetic field. This distribution of site poles suggests that geomagnetic secular variation during the Late Triassic consisted mainly of dipole wobble, with little or no contribution from nondipole sources. No reversals of the field are recorded in Upper Triassic igneous rocks from eastern North America, although reversals are known elsewhere in the Triassic section. Available evidence indicates that, although North America was actively drifting during the Late Triassic and Jurassic, polar wandering during that time interval was minimal. A model based on work by E. Irving, A. Cox, P. Goldreich, and A. Toomre is used to predict a correlation between general convective overturn in the upper mantle, polar wandering, high intensities of the nondipole component of the geomagnetic field, and high frequencies of geomagnetic reversal.

Beck [1965] described paleomagnetic results for 20 sampling sites located in three Late Triassic diabase intrusions in southeastern Pennsylvania. The present report expands the number of useful sites to 78 (of 105 originally sampled) and the number of intrusions studied to four. The addition of extra data does not materially change the paleomagnetic results of the earlier study, although it does improve precision. However, the new data do permit us to test alternative descriptions of the dispersion of magnetic directions and paleomagnetic poles. Such a test should contribute to our understanding of the nature of the geomagnetic secular variation. For this reason, the full set of Late Triassic data are described here in detail.

The geology and tectonic history of the Triassic basins of eastern North America have been discussed by *Sanders* [1963], *de Boer* [1968], and *Klein* [1969], among others. In the Gettysburg basin of southeastern Pennsylvania, four major diabase bodies intrude a thick section of continental clastic sedimentary rock (Figure 1). The diabase intrusions for the most part are concordant and sheetlike in form, ranging up to 1000 meters in thickness; crosscutting and irregular masses are present locally. Some of the thicker bodies show evidence of gravity fractionation along general lines described by *Walker* [1940] for the Palisade diabase sill; magnetic intensity and stability data indicate that the magnetic oxide minerals also vary systematically with position in the sill [*Beck*, 1966]. Agreement between directions of remanent magnetization of sedimentary, volcanic, and intrusive rocks in the Newark and Gettysburg basin [*Opdyke*, 1961; *Beck*, 1965] indicates that the intrusions were emplaced and magnetized shortly after deposition of the sedimentary material, but before any significant amount of tilting of the layered rocks had taken place.

Figure 1 shows the major diabase intrusions in southeastern Pennsylvania and the outline of the Gettysburg basin. Layered rocks within the basin dip gently to the north and northwest. Attitudes appropriate to each sampling site are listed in Table 1. These strikes and dips represent rotations of the diabase bodies after magnetization and have been corrected for paleomagnetic analysis.

The Late Triassic age assigned to the Gettysburg basin rock is supported by faunal evidence [*Willard et al.*, 1959] and by the radiometric

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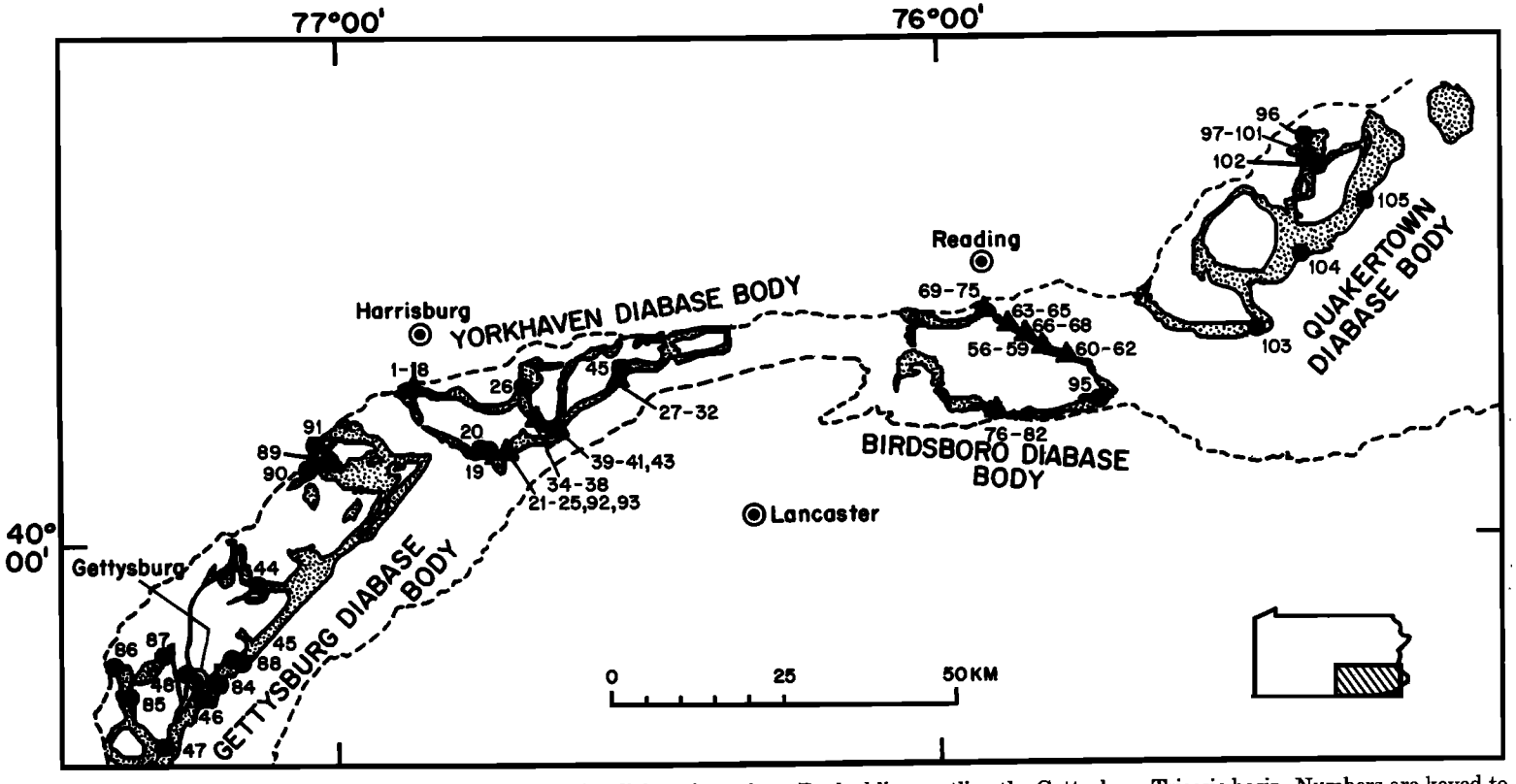


Fig. 1. Location of sites. Stippled pattern indicates major diabase intrusions. Dashed lines outline the Gettysburg Triassic basin. Numbers are keyed to Table 1 and identify individual sites. A triangular location mark indicates places where continuous or near-continuous exposures are available through a major thickness of a single intrusion. At such localities, more than one site was established. Map modified from the geological map of Pennsylvania.

TABLE 1. Directions of Magnetization, Dispersion, and Attitudes for Upper Triassic Diabase

			NRM, In Situ					150-oe, Corrected						
S	*N*	Location	*D*	*I*	κ	α_{95}	δ	*D*	*I*	κ	α_{95}	δ	Strike	Dip
						Gettysburg Diabase Sheet								
44	4	39.95°N, 77.13°W	12.5	38.5	28.5	17.5	13.1	13.0	32.5	28.4	17.5	13.2	N50°E	25°N
45	5	39.85°N, 77.17°W	12.0	37.5	451.2	3.6	3.4	2.0	23.0	335.0	4.2	4.0	N50°E	20°N
46	4	39.80°N, 77.21°W	16.0	36.5	385.8	4.7	3.6	1.5	31.0	1043.2	2.9	2.2	N25°E	20°W
47	5	39.74°N, 77.28°W	2.5	27.5	26.3	15.2	14.1	358.5	16.5	61.0	9.9	9.3	N40°E	25°W
48	2		*											
84	4	39.85°N, 77.16°W	16.0	34.5	369.8	4.8	3.7	6.5	23.0	610.8	3.7	2.8	N50°E	20°N
85	5	39.80°N, 77.34°W	19.0	47.5	12.2	22.9	20.8	11.0	30.0	18.7	18.2	16.8	N55°E	15°N
86	5		†											
87	3		*											
88	5	39.83°N, 77.19°W	19.5	33.5	2063.9	1.7	1.6	8.5	29.5	1228.9	2.2	2.1	N25°E	20°W
89	5	40.11°N, 77.01°W	352.5	68.5	12.3	22.7	20.7	347.0	51.5	37.0	12.7	11.9	N40°E	20°W
90	5	40.10°N, 77.04°W	337.5	44.0	26.0	15.3	14.2	14.5	14.0	11.9	23.1	21.0	N40°E	20°W
91	5	40.12°N, 77.03°W	355.5	24.0	47.5	11.2	10.5	353.5	11.0	164.3	6.0	5.7	N55°E	15°N
						Yorkhaven Diabase Sheet								
1	2	40.20°N, 76.87°W	*											
2	4	40.20°N, 76.87°W	3.0	43.0	10.9	29.2	21.3	353.5	9.0	154.2	7.4	5.7	N80°E	35°N
3	4	40.20°N, 76.87°W	33.0	28.0	2.6	73.3	43.9	1.5	16.5	106.0	9.0	6.8	N80°E	35°N
4	4	40.20°N, 76.87°W	11.0	62.0	157.5	7.3	5.6	2.0	30.0	92.7	9.6	7.3	N80°E	35°N
5	4	40.20°N, 76.87°W	5.5	67.0	219.4	6.2	4.7	3.0	25.5	59.9	12.0	9.1	N80°E	35°N
6	4	40.20°N, 76.87°W	15.0	65.5	90.3	9.7	7.4	3.0	31.5	87.1	9.9	7.5	N80°E	35°N
7	4	40.20°N, 76.87°W	1.5	59.5	63.3	11.6	8.8	357.5	21.5	51.4	12.9	9.8	N80°E	35°N
8	5	40.20°N, 76.87°W	355.5	58.5	5.8	34.8	30.1	1.0	16.0	61.0	9.9	9.3	N80°E	35°N
9	2		*											
10	5	40.20°N, 76.87°W	6.0	50.5	44.5	11.6	10.9	352.0	10.5	59.9	10.0	9.4	N80°E	35°N
11	5	40.20°N, 76.87°W	19.0	59.0	179.4	5.7	5.4	1.0	12.5	114.6	7.2	6.8	N80°E	35°N
12	5	40.20°N, 76.87°W	33.0	53.5	5.8	34.9	30.2	2.5	19.5	34.2	13.3	12.4	N80°E	35°N
13	5	40.20°N, 76.87°W	353.0	34.0	7.0	31.2	27.5	5.0	14.5	177.0	5.8	5.5	N80°E	35°N
14	5	40.20°N, 76.87°W	10.0	34.5	3.1	52.0	41.0	356.5	21.5	52.0	10.7	10.1	N80°E	35°N
15	4	40.20°N, 76.87°W	11.0	30.0	4.4	49.6	33.5	3.5	19.0	20.1	21.0	15.7	N80°E	35°N
16	5	40.20°N, 76.87°W	11.5	49.5	13.9	21.3	19.5	0.0	18.5	38.1	12.6	11.7	N80°E	35°N
17	4	40.20°N, 76.87°W	38.5	2.5	3.3	49.6	39.7	8.5	7.5	577.7	3.8	2.9	N80°E	35°N

TABLE 1. (continued)

S	N	Location	NRM, In Situ D	I	κ	α_{95}	δ	150-oe, Corrected D	I	κ	α_{95}	δ	Strike	Dip
					Yorkhaven Diabase Sheet									
18	2		*											
19	5	40.12°N, 76.75°W	1.0	46.5	356.7	4.1	3.8	6.5	27.5	27.6	14.8	13.8	N40°E	20°W
20	5	40.12°N, 76.75°W	8.5	40.0	385.9	3.9	3.7	358.0	26.5	686.7	2.9	2.8	N40°E	20°W
21	5	40.12°N, 76.72°W	355.0	48.5	43.5	11.7	11.0	359.5	26.0	283.6	4.6	4.3	N40°E	20°W
22	5	40.12°N, 76.72°W	353.5	56.0	7.6	29.8	26.4	358.5	27.0	101.3	7.6	7.2	N40°E	20°W
23	5	40.12°N, 76.72°W	349.0	39.5	6.5	32.4	28.4	351.0	24.0	9.4	26.3	23.8	N40°E	20°W
24	5	40.12°N, 76.72°W	350.0	55.5	14.1	21.1	19.3	351.0	35.5	20.8	17.2	16.0	N40°E	20°W
25	5	40.13°N, 76.72°W	2.0	53.5	37.1	12.7	11.9	359.0	28.0	169.6	5.9	5.6	N40°E	20°W
26	4	40.21°N, 76.69°W	359.5	44.0	183.1	6.8	5.2	357.5	16.5	189.1	6.7	5.1	N65°E	25°N
27	4	40.21°N, 76.57°W	13.0	42.0	89.2	9.8	7.4	359.0	16.0	67.5	11.3	8.5	N40°E	35°W
28	5	40.21°N, 76.57°W	359.5	42.0	48.7	11.1	10.4	358.0	17.5	128.6	6.8	6.4	N40°E	35°W
29	5	40.21°N, 76.57°W	10.0	37.5	258.1	4.8	4.5	0.5	16.0	1171.7	2.2	2.1	N40°E	35°W
30	5	40.21°N, 76.57°W	8.5	38.0	43.3	11.8	11.0	355.5	18.0	1042.4	2.4	2.2	N40°E	35°W
31	4	40.21°N, 76.57°W	358.0	35.5	25.3	18.6	14.0	356.0	17.0	779.0	3.3	2.5	N40°E	35°W
32	5	40.21°N, 76.57°W	9.0	42.0	98.3	7.8	7.3	354.0	19.5	99.4	7.7	7.3	N40°E	35°W
33	4	40.21°N, 76.57°W	§											
34	5	40.16°N, 76.67°W	17.0	49.5	1350.7	2.1	2.0	359.0	31.5	1157.4	2.3	2.1	N45°E	30°NW
35	5	40.16°N, 76.67°W	18.0	43.0	29.1	14.4	13.4	2.5	24.0	26.1	15.3	14.2	N45°E	30°NW
36	4	40.16°N, 76.67°W	26.5	49.5	44.2	14.0	10.6	5.0	33.0	42.7	14.2	10.7	N45°E	30°NW
37	5	40.16°N, 76.67°W	8.0	48.0	12.5	22.5	20.5	2.0	25.5	306.1	4.4	4.1	N45°E	30°NW
38	5	40.16°N, 76.67°W	16.0	50.0	123.7	6.9	6.5	357.0	28.5	114.0	7.2	6.8	N45°E	30°NW
39	4	40.15°N, 76.64°W	5.0	43.5	238.2	5.0	4.7	354.0	19.5	387.5	3.9	3.7	N50°E	30°N
40	5	40.15°N, 76.65°W	4.0	46.0	617.9	3.1	2.9	352.5	20.5	767.3	2.8	2.6	N50°E	30°N
41	5	40.15°N, 76.63°W	14.5	37.0	40.6	12.2	11.4	0.0	20.0	830.6	2.7	2.5	N50°E	30°N
42	5	40.23°N, 76.54°W	8.0	1.5	5.0	38.3	32.6	354.0	16.5	244.4	4.9	4.6	N40°E	35°W
43	5	40.15°N, 76.63°W	14.0	43.0	123.6	6.9	6.5	358.5	21.5	610.4	3.1	2.9	N50°E	30°N
92	5	40.13°N, 76.72°W	356.5	38.5	128.7	6.8	6.4	358.0	24.0	1069.7	2.3	2.2	N40°E	20°W
93	5	40.13°N, 76.72°W	0.0	39.5	718.4	2.9	2.7	354.5	24.0	731.9	2.8	2.7	N40°E	20°W

TABLE 1. (continued)

S	N	Location	NRM, In Situ D	NRM, In Situ I	NRM, In Situ κ	NRM, In Situ α_{95}	NRM, In Situ δ	150-oe, Corrected D	150-oe, Corrected I	150-oe, Corrected κ	150-oe, Corrected α_{95}	150-oe, Corrected δ	Strike	Dip
						Birdsboro Diabase Sheet								
50	5	40.25°N, 75.82°W	335.0	58.5	21.4	16.9	15.7	353.5	34.0	44.7	11.6	10.8	N75°E	25°N
51	5	40.25°N, 75.82°W	14.5	63.0	97.9	7.8	7.3	4.0	34.5	67.4	9.4	8.8	N75°E	25°N
52	5	40.25°N, 75.82°W	354.0	56.5	21.4	16.9	15.7	356.5	29.0	76.7	8.8	8.3	N75°E	25°N
53	5	40.25°N, 75.82°W	35.5	49.0	13.7	21.4	19.6	11.0	27.0	34.2	13.3	12.4	N75°E	25°N
54	5	40.25°N, 75.82°W	18.0	-16.5	2.0	74.3	51.0	6.5	15.5	53.6	10.5	9.9	N75°E	25°N
55	5	40.25°N, 75.82°W	22.5	38.0	15.4	20.1	18.5	357.5	12.5	29.8	14.2	13.3	N75°E	25°N
56	5	40.25°N, 75.82°W	3.0	48.0	20.4	17.3	16.0	4.5	8.5	49.1	11.0	10.4	N75°E	25°N
57	5	40.25°N, 75.82°W	6.5	52.0	12.1	22.9	20.8	358.0	27.5	182.8	5.7	5.4	N75°E	25°N
58	5	40.25°N, 75.82°W	14.0	42.0	1.3	138.0	63.3	352.5	34.5	32.7	13.6	12.7	N75°E	25°N
59	5	40.25°N, 75.82°W	28.5	58.0	4.1	43.4	36.0	358.5	34.0	49.6	11.0	10.3	N75°E	25°N
60	5	40.24°N, 75.78°W	22.5	32.5	59.5	10.0	9.4	14.5	11.0	166.0	6.0	5.6	N80°E	25°N
61	5	40.24°N, 75.78°W	27.5	42.0	10.0	25.4	22.9	10.0	19.5	71.5	9.1	8.6	N80°E	25°N
62	5	40.24°N, 75.78°W	356.0	29.0	21.1	17.0	15.8	3.5	16.5	282.4	4.6	4.3	N80°E	25°N
63	4	40.28°N, 75.88°W	337.0	27.0	1.9	80.0	52.8	349.0	40.0	174.0	7.0	5.3	N35°E	20°W
64	4	40.28°N, 75.88°W	14.5	35.5	14.4	25.1	18.5	355.0	30.0	31.9	16.5	12.4	N35°E	20°W
65	4	40.28°N, 75.88°W	350.5	24.0	5.8	42.0	29.2	356.5	43.0	363.4	4.8	3.7	N35°E	20°W
66	4	40.26°N, 75.85°W	343.5	48.0	21.5	20.3	15.1	354.0	33.0	60.2	11.9	9.0	N40°E	20°W
67	5	40.26°N, 75.85°W	13.5	55.5	5.3	35.3	30.5	346.0	25.0	18.4	15.6	13.6	N40°E	20°W
68	4	40.26°N, 75.85°W	39.5	61.5	1.3	160.7	64.6	346.5	27.5	86.2	10.0	7.6	N40°E	20°W
69	5	40.30°N, 75.92°W	96.0	31.0	3.0	53.1	41.6	357.5	5.5	11.3	23.8	21.6	E-W	40°N
70	4	40.30°N, 75.92°W	1.0	54.0	9.7	31.2	22.6	359.5	17.5	100.3	9.2	7.0	E-W	40°N
71	5	40.30°N, 75.92°W	4.5	62.5	2.9	54.5	42.4	358.0	27.0	17.7	18.7	17.2	E-W	40°N
72	5	40.30°N, 75.92°W	232.5	38.5	1.3	161.6	64.6	36.0	27.5	2.2	68.7	48.8	E-W	40°N
73	5	40.30°N, 75.92°W	9.0	65.5	6.3	33.2	29.0	356.0	12.0	66.0	9.5	8.9	E-W	40°N

TABLE 1. (continued)

S	N	Location	NRM, In Situ D	NRM, In Situ I	NRM, In Situ κ	NRM, In Situ α_{95}	NRM, In Situ δ	150-oe, Corrected D	150-oe, Corrected I	150-oe, Corrected κ	150-oe, Corrected α_{95}	150-oe, Corrected δ	Strike	Dip
						Birdsboro Diabase Sheet								
74	4	40.30°N, 75.92°W	158.5	81.0	1.4	155.3	58.8	359.5	18.0	71.3	11.0	8.3	E-W	40°N
75	5	40.30°N, 75.92°W	50.0	54.0	2.7	57.7	44.0	351.5	-7.0	19.4	17.8	16.5	E-W	40°N
76	5	40.17°N, 75.91°W	6.5	40.5	328.8	4.2	4.0	359.0	12.5	330.6	4.2	4.0	N70°E	30°N
77	5	40.26°N, 75.85°W	317.0	10.5	2.2	67.5	48.4	325.0	30.0	9.1	26.8	24.0	N40°E	20°W
78	5	40.26°N, 75.85°W	354.0	33.0	4.4	41.4	34.7	351.5	33.5	14.8	20.6	18.9	N40°E	20°W
79	5	40.17°N, 75.91°W	13.0	44.0	75.9	8.8	8.3	1.0	12.0	199.2	5.4	5.1	N70°E	30°N
80	5	40.17°N, 75.91°W	18.5	48.5	25.3	15.5	14.4	358.0	19.0	25.3	15.5	14.4	N70°E	30°N
81	5	40.17°N, 75.91°W	32.0	62.5	6.1	33.7	29.3	354.5	21.0	164.9	6.0	5.6	N70°E	30°N
82	5	40.17°N, 75.91°W	31.5	30.5	6.9	31.5	27.7	1.0	9.5	76.5	8.8	8.3	N70°E	30°N
94	5		¶											
95	5	40.18°N, 75.73°W	10.0	37.5	177.1	5.8	5.5	1.0	29.0	819.9	2.7	2.5	N55°E	15°N
						Quakertown Diabase Sheet								
96	5	40.50°N, 75.38°W	315.5	19.5	12.6	22.4	20.4	3.5	22.0	128.5	6.8	6.4	N35°E	15°W
97	5	40.50°N, 75.38°W	39.0	7.5	8.0	28.7	25.6	9.0	30.0	79.6	8.6	8.1	N35°E	15°W
98	5	40.49°N, 75.37°W	347.5	33.5	5.3	36.6	31.4	1.0	24.5	141.4	6.5	6.1	N35°E	15°W
99	5	40.49°N, 75.37°W	325.0	23.5	13.3	21.7	19.9	358.5	28.0	131.3	6.7	6.3	N35°E	15°W
100	5	40.48°N, 75.36°W	51.0	13.0	9.4	26.3	23.6	6.0	26.0	234.8	5.0	4.7	N35°E	15°W
101	5	40.48°N, 75.36°W	30.5	20.0	162.9	6.0	5.7	7.0	25.0	1464.6	2.0	1.9	N35°E	15°W
102	5	40.47°N, 75.35°W	27.0	35.0	28.6	14.6	13.6	356.5	47.0	107.9	7.4	7.0		
103	5	40.28°N, 75.47°W	9.5	49.5	29.0	14.5	13.5	8.0	40.0	25.7	15.4	14.3	N70°E	10°N
104	5	40.36°N, 75.32°W	10.0	47.0	222.0	5.2	4.9	1.0	34.0	159.0	6.1	5.8	N50°E	15°N
105	5	40.43°N, 75.28°W	6.0	44.5	175.1	5.8	5.5	359.0	39.0	166.3	6.0	5.6	N40°E	10°W

S, site designation; *N*, number of samples; *D* and *I*, mean declination and inclination of site; κ, estimate of Fisher's precision parameter [*Fisher*, 1953]; α_{95}, radius of cone of confidence, 95% probability level; δ, angular dispersion factor [*Wilson*, 1959]; strike and dip, strike and dip of host strata. Natural remanent magnetization (NRM) directions have not been corrected for attitude of host strata, whereas magnetically cleaned directions (150-oe peak alternating field) have been so corrected.

*Too few samples.

†Badly scattered; field evidence of shearing and alteration.

§Orientations untrustworthy because of large and variable declination anomalies.

¶Streaked distribution.

ages of related igneous rocks in basins to the northeast. Potassium-argon ages summarized by *de Boer* [1968], *Carmichael and Palmer* [1968], and *Larochelle and Wanless* [1966] range roughly from 190 to 200 m.y. This is Late Triassic, according to the Phanerozoic time scale of the Geological Society of London.

PALEOMAGNETIC RESULTS

Figure 1 shows the distribution of sampling sites. From 3 to 6 oriented samples (field-drilled cores supplemented by hand samples) were taken at each site. A magnetic compass, checked by the back-azimuth technique, was used for orientation. Sites were located exclusively in fresh artificial exposures, and the samples collected at each site were spread over distances ranging from 5 to 25 meters. Remanent magnetization of a specimen from each sample was measured on a spinner magnetometer [*Doell and Cox*, 1965], and magnetic susceptibility was measured on a standard inductance bridge [*Anderson*, 1961]. Magnetic intensity data for these rocks have been reported previously [*Beck*, 1966]. All specimens were magnetically cleaned by the alternating-field method at a peak value of 150 oe. The value of 150 oe was chosen by using the minimum-dispersion criterion of *Irving et al.* [1961] after progressive demagnetization of 24 randomly selected sites at 37.5-, 75-, 150-, and 300-oe peak fields. Most sites showed considerable stability, but many experienced an increase in dispersion over 150 oe. A few sites were demagnetized beyond 300 oe, but no significant change in mean direction nor reduction in scatter was found.

Directions of natural remanent magnetization (NRM) and directions after magnetic cleaning for the four diabase intrusions are summarized in Table 1. The NRM directions in Table 1 have not been corrected for geological dip; consequently, these directions are directly suitable for analysis of magnetic survey data.

Not all sites studied yielded results of acceptable accuracy. In order to insure high reliability for the paleomagnetic results and to minimize dispersion not related to the true dispersion of the ancient geomagnetic field, the following rejection criteria were applied. (1) At least four samples had to survive the sampling and specimen-preparation processes. (2) Streaked (markedly elongate) distributions were rejected on the grounds that some secondary magnetization might have survived the magnetic cleaning. Only one site was so rejected. (3) Sites for which the circle of 95% confidence [*Fisher*, 1953] exceeded 15° in radius were rejected as giving an unacceptably imprecise measure of the direction of the ancient geomagnetic field. The value of 15° is completely arbitrary; *Wilson* [1970], for instance, uses 12°, and *Irving* [1964] rejects results for which α_{95} exceeds 25°. Slightly more than ¼ of the sites were rejected using these three criteria.

Average directions of remanent magnetization and paleomagnetic pole positions have been calculated for each diabase intrusion and are given in the first four rows of Table 2. The

TABLE 2. Mean Directions of Magnetization and Paleomagnetic Pole Positions for Upper Triassic Diabase Intrusions, Southeastern Pennsylvania

		Direction					Poles		
Intrusion	B	D, deg	I, deg	κ	α_{95}, deg	δ	Location	κ'	α_{95}', deg
Gettysburg	7	0.0	26.5	32.2	11.0	13.0	65.0°N, 103.0°E	53.9	8.3
Yorkhaven	37	359.0	21.0	121.4	2.2	7.2	61.0°N, 105.5°E	236.6	1.5
Birdsboro	26	359.5	23.5	46.0	4.3	11.7	62.0°N, 105.0°E	85.1	3.2
Quakertown	9	2.5	30.5	84.4	5.6	8.3	66.0°N, 98.5°E	143.6	4.3
All sites	78						62.0°N, 104.5°E	118.1	1.5
All intrusions	4						63.5°N, 103.0°E	839.0	3.2

B, number of sites; D and I, mean declination and inclination; κ, α_{95}, and δ, as explained for Table 1 (primed values were calculated by using site poles as data, whereas unprimed values were calculated by using site mean directions as data). All samples magnetically cleaned in 150-oe peak field.

pole reported for each intrusion is the mean of poles for individual sampling sites and was not calculated from the intrusion's mean direction of remanent magnetization. Poles rather than magnetization directions are used throughout this study to facilitate comparison with paleomagnetic results from correlative rock units from other Triassic basins. Geomagnetic directions for the Late Triassic dipole field in southeastern Pennsylvania change by less than 1.5° in 100 km. Thus, site mean directions can be compared directly within intrusions, but average directions between intrusions and between rock units from separate Triassic basins are better compared by means of paleomagnetic poles.

The fifth row in Table 2 gives a Late Triassic paleomagnetic pole for diabase intrusions from southeastern Pennsylvania. This pole is the mean of site poles for the 78 sites that met all reliability criteria. As each site probably provides an independent spot observation of the Late Triassic geomagnetic field, this mean of site poles should be the best estimate available from the data of this study of the position of the Late Triassic paleomagnetic pole. As such, it replaces the pole reported by *Beck* [1965], from which it differs by less than 0.5°. A second pole calculation, using the mean poles for each intrusion as independent observations, is shown in the sixth row of Table 2. The low dispersion of this calculation probably results from the fact that the poles for the four separate intrusions are not independent estimates of pole position, but rather represent the means of four separate samples drawn from the same population of directions. A similar low value of dispersion was found when comparing poles for the separate lithological units that comprise the Duluth gabbro complex [*Beck*, 1970].

DISPERSION OF DIRECTIONS AND POLES

Directions of remanent magnetization at different sites within a single rock body always show a certain amount of scatter; part of this scatter is related to experimental difficulties of many sorts (*Creer* [1962] provides a short summary), but the rest reflects actual dispersion of the ancient geomagnetic field. Under favorable circumstances, experimental error can be reduced to insignificant levels, as in the present study, by careful selection of sampling sites, the use of systematic magnetic cleaning, and by applying fairly exacting rejection criteria. Probably the only major experimental source of dispersion contributing to the scatter of directions in the present study is error in selecting the proper tectonic correction. Even here, however, errors are likely to be of the order of 5° or less and randomly directed. The dispersion of site mean directions from Table 2 thus should mainly reflect actual changes in the direction of the Late Triassic geomagnetic field.

The exact range of geomagnetic history sampled during the cooling of the four diabase sheets is uncertain, but probably is of the order of 10^4 to 10^5 years. Individual sheets range up to 1000 meters in thickness, and from Figures 2 and 3 of *Jaeger* [1957, pp. 312, 313] it appears that a sheet of this thickness would acquire its TRM over a period of approximately 7000 years. *Clark* [1969] estimated 2000 to 6000 years for the Marys Peak sill, Oregon, which is 390 meters thick. Thus 7000 years can serve as a minimum estimate; the actual time probably is much greater than this, as it is unlikely that intrusive sheets emplaced at several stratigraphic horizons and having an extent along strike of more than 200 km would begin to cool through their blocking temperatures everywhere at exactly the same time. Moreover, the fact that the paleomagnetic pole calculated from this study agrees almost exactly with the pole calculated by *Opdyke* [1961] for Upper Triassic diabase, basalt, and a considerable thickness of sedimentary rock in New Jersey and eastern Pennsylvania argues that the time span represented by the Pennsylvania diabase sheets was long relative to the time span of the geomagnetic secular variation.

Dispersion of directions within the four diabase intrusions is quite low relative to the well-known Creer-Cox model, in which the latitude-dependence of scatter caused by geomagnetic secular variation is estimated from scatter of the present field, averaged over small circles of latitude [*Creer*, 1962; *Cox*, 1962]. For instance, *Creer* [1962, Figure 6, p. 3472] predicts an angular dispersion factor of about 17° for the paleolatitude of southeastern Pennsylvania diabase sills (12°N). As is shown in Table 2, angular dispersion for the four diabase intrusions ranges from 7.2° to 13.0°. All but

the Gettysburg intrusion show angular dispersion factors that are significantly lower (95% probability [*Cox*, 1969*a*]) than the value of 17° predicted by Creer's model. Moreover, the angular dispersion factor of all 78 site mean directions averaged together (7.4°) is significantly lower than the predicted value, even though slightly augmented by inclination differences reflecting differences in paleolatitude. Evidently geomagnetic secular variation was significantly less during at least part of the Late Triassic than it is at present. Since studies by *Creer and Sanver* [1969] suggest that secular variation during the Quaternary and Late Tertiary was slightly greater on the average than at present, the sampled part of the Late Triassic geomagnetic field appears definitely to have been anomalous.

It might be argued that the low dispersion of site mean directions and poles noted above and attributed to low geomagnetic secular variation in the Late Triassic was instead the result of a very prolonged magnetization process, whereby much of the secular variation was averaged out in the within-site dispersion. By this argument, mean directions for sampling sites do not represent points in time, and their dispersion is less than the dispersion inherent in the secular variation. *Storetvedt* [1968] has suggested that this must be true in general for igneous bodies subject to very slow cooling in which, he reasons, the stable remanence is likely to have been acquired slowly at low temperatures by a process akin to the acquisition of VRM. In support of this idea, Storetvedt cites examples of groups of lavas and dikes for which within-site and between-site dispersion are equal. This is not the case for the Pennsylvania Triassic; within-site dispersion is generally much less than between-site dispersion (Table 1). Apparently the cooling and magnetization processes were much less complex in the Pennsylvania sills than in the examples cited by Storetvedt.

To test for artificial smoothing of the secular variation, a calculation was run using only site mean directions for sites for which Fisher's precision parameter [*Fisher*, 1953] exceeded 200. The angular dispersion for these sites, 26 in all, ranged from 2.1° to 4.7°; it would be difficult to argue that the within-site dispersion for such tightly grouped sites represents much more than experimental error. The mean of site poles for this calculation lies at 62°N, 103.5°E, with precision parameter 171.2 and angular dispersion 6.1°. Dispersion for this calculation is less than the dispersion for all 78 sites (fifth row, Table 2), not greater, as would be expected if some of the true dispersion of the geomagnetic field had been averaged out within the high-dispersion sites.

Secular variation of the directional elements of the geomagnetic field might be caused by wobble of the dipole axis, by variation in dipole intensity, by drift and change of the nondipole component of the field, or by a combination of several causes. If dipole wobble is the only source of paleosecular variation, and if the wobble is random, the distribution of site mean poles should have circular symmetry about the mean pole position. Alternatively, if secular variation is caused by random perturbations of the dipole by the nondipole field, site mean directions should be distributed with circular symmetry. In general, a set of poles distributed with circular symmetry on a sphere will map into an oval distribution of magnetic field directions, and vice versa. If random dipole wobble and random perturbation of the dipole direction by the nondipole field are both active and of roughly equal importance, in general neither poles nor directions will show circular symmetry. Finally, it is possible that one or both sources of secular variation will show directional bias. Bias of apparent dipole motion, for instance, might be produced by significant polar wandering during the time in which a particular rock unit acquired its remanent magnetization. This process would produce an approximately oval distribution of poles, with the long axis of the distribution in the direction of polar motion. The distribution of poles and field directions under different assumed conditions of behavior of the secular variation is discussed by *Cox* [1970] and by *Brock* [1971].

Figure 2 is a plot of poles for sites in Upper Triassic diabase of southeastern Pennsylvania; to eliminate visual distortion inherent in the projection technique, the distribution has been rotated by a standard coordinate transformation until its mean coincides with the pole of projection. There appears to be a slight elongation of the distribution in the second and fourth quadrants, but this is more apparent than real;

a χ^2 calculation (Table 3) indicates that the poles conform to a circularly symmetrical *Fisher* [1953] distribution (with κ of 118.1) to a high degree of probability. A plot of field directions (not shown), similarly rotated, is highly elongate in the direction of a great circle connecting mean pole position and sampling site. This is the predicted result of the pole-to-field direction mapping relationship operating on a circular distribution of poles. A χ^2 goodness-of-fit calculation (Table 3) rejects (95% confidence) the hypothesis that the distribution of field directions is Fisherian.

A circular distribution of poles satisfying Fisher's distribution could come about in several ways. For instance, secular variation during the Late Triassic may have consisted solely of a slight wobble of the geomagnetic dipole, resembling model B of *Creer et al.* [1959] (see also *Irving* [1964, p. 50] and *Cox* [1970]), with an angular standard deviation of about 7.5°. Alternatively, *Cox* [1970] has noticed, with regard to the present magnetic field, that the dispersion of field directions near the equator that would be produced by rotating its nondipole component is asymmetrical and non-Fisherian, but would produce an approximately Fisherian distribution of poles. Perhaps dispersion of the sort described in this study is characteristic of units that acquire their remanent magnetism at low latitude. In this regard, it may be significant that the Upper Carboniferous Great Whin sill [*Creer et al.*, 1959]

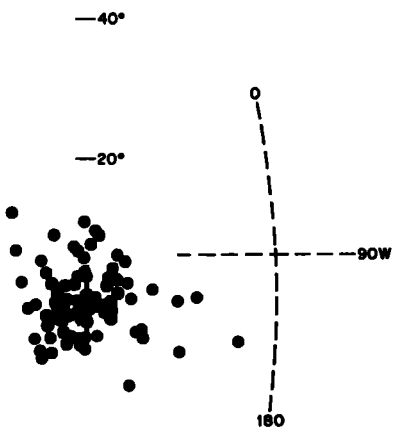


Fig. 2. Site poles plotted with their mean at the center of equal-area projection. Dashed lines are geographic meridians.

TABLE 3. χ^2 Calculation for Goodness of Fit of Poles and Site Mean Directions to a Fisher Distribution

	δ for Poles		δ for Directions	
D'	<6.2°	>6.2°	<8.5°	>8.5°
0 to 71	7	7	8	8
72 to 143	12	7	8	3
144 to 215	10	5	12	14
216 to 287	8	8	2	1
288 to 359	8	6	15	7
	7.8 expected in each interval			
	χ^2 = 4.86		χ^2 = 27.14	

D', declination of pole or site mean direction after coordinate transformation placing mean at center of projection; δ, angle between pole or site mean direction and mean pole or mean of site mean directions; χ^2, the value computed by

$$\chi^2 = \sum_{i=1}^{10} \frac{(f_i - 7.8)^2}{7.8}.$$

also was magnetized near the equator, and it too has a Fisherian distribution of site poles with a corresponding oval distribution of magnetic field directions (however, see *Storetvedt and Gidskehaug* [1969] for additional discussion of the magnetization of the Great Whin sill). On the other hand, site poles in the Keweenawan Duluth gabbro complex, which was magnetized at a latitude of about 35°, are strongly asymmetric [*Beck*, 1970, and unpublished data]. For the Great Whin sill and the southeastern Pennsylvania diabase intrusions, dispersion (directions or poles) is significantly lower than predicted by the present-field model, but the same is not true for the Duluth gabbro. Thus it seems possible that the Great Whin sill and Pennsylvania diabase intrusions show the effect of dipole wobble acting alone, with little contribution from a nondipole component of the geomagnetic field.

It would be valuable, in regard to a general study of the geomagnetic secular variation, to examine the dispersion characteristics of very precisely correlative rock units from a wide range of paleolatitude. The difficulty here appears to lie in finding an adequate correlation technique. Using rocks with similar radiometric ages from several parts of the world is not feasible, because it is by no means certain that the character of the geomagnetic secular variation is constant over time periods comparable to the usual limits of error in radiometric

dating. What is needed is a correlation technique based on the magnetic field itself. *Cox* [1969*b*] has used such a technique in studying secular variation during the Brunhes normal polarity epoch. For older rock units, precise correlation is harder to establish. It might, however, be possible to recognize the beginning or end of major polarity intervals, such as the late Paleozoic reversed interval, on a worldwide basis, and thus establish very precise correlation between widely scattered rock units.

DISCUSSION

The paleomagnetism of rocks from the Upper Triassic basins of eastern North America has become one of the more thoroughly explored topics of research in rock magnetism. At least eight separate studies are involved; current summaries are provided by *de Boer* [1968] and *Carmichael and Palmer* [1968]. All remanence directions measured to date show normal (negative) polarity, if the interpretation placed by *de Boer* [1968] on results of *Kobayashi and Schwarz* [1966] is correct. In view of the considerable thickness and geographic extent of Upper Triassic rocks involved in these studies, it seems reasonable to assume that they represent a time period of the order of 10^6 years or more, and therefore that the geomagnetic field at some point during the Late Triassic maintained a constant polarity for a roughly comparable period. This observation contrasts with results given by *Picard* [1964] and C. E. Helsley (personal communication, 1972) showing frequent reversals at other times during the Triassic, and with analyses of marine magnetic anomalies suggesting an average duration of about 4 to 5 $\times$ 10^5 years for polarity intervals during the Tertiary [*Heirtzler et al.*, 1968]. The mean duration of polarity intervals calculated by *Cox* [1968] for the past 10 m.y. is 2.2 $\times$ 10^5 years. Although reversal frequency throughout much of the Mesozoic seems to have been roughly comparable to reversal frequency during the Tertiary (C. E. Helsley, personal communication, 1972), long periods of normal polarity are known from the Cretaceous [*Helsley and Steiner*, 1969].

If, as *Cox* [1968, 1969*c*] suggested, there is a causal relationship between a high ratio of nondipole-to-dipole field intensity and a high frequency of reversal of the geomagnetic field, one would expect infrequent reversals during the period when the southeastern Pennsylvania diabase intrusions became magnetized, since the nondipole field intensity at that time appears to have been relatively low. The Upper Triassic results thus are consistent with Cox's model. However, it is possible that the Upper Triassic rocks of eastern North America merely represent a brief period of time, of the order of 5 $\times$ 10^5 years or less, and thus tell us little about the length of geomagnetic polarity epochs during the Triassic. This is considered unlikely in view of the polar wandering observations discussed below.

An attempt to compare paleomagnetic poles for various parts of the Upper Triassic belt of eastern North America is shown in Figure 3, in which the distribution again has been rotated to the center of the projection (the same data are shown unrotated in Figure 4).

Figures 3 and 4 are not a wholly satisfactory representation of the available paleomagnetic data, but, in view of the variety of statistical techniques used, they are perhaps the best that

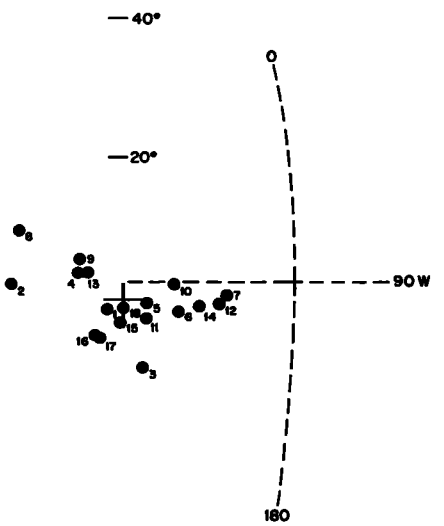


Fig. 3. Poles for igneous rock bodies (large intrusions, contiguous groups of small intrusions, sequences of lava flows), plotted with their mean at the center of an equal-area projection: (1, 2) recalculated from *Irving and Banks* [1961]; (3, 4) recalculated from *Opdyke* [1961]; (5) from *Larochelle and Wanless* [1966]; (6, 7) from *Carmichael and Palmer* [1968]; (8–14) from *de Boer* [1968]; (15–18) from this work.

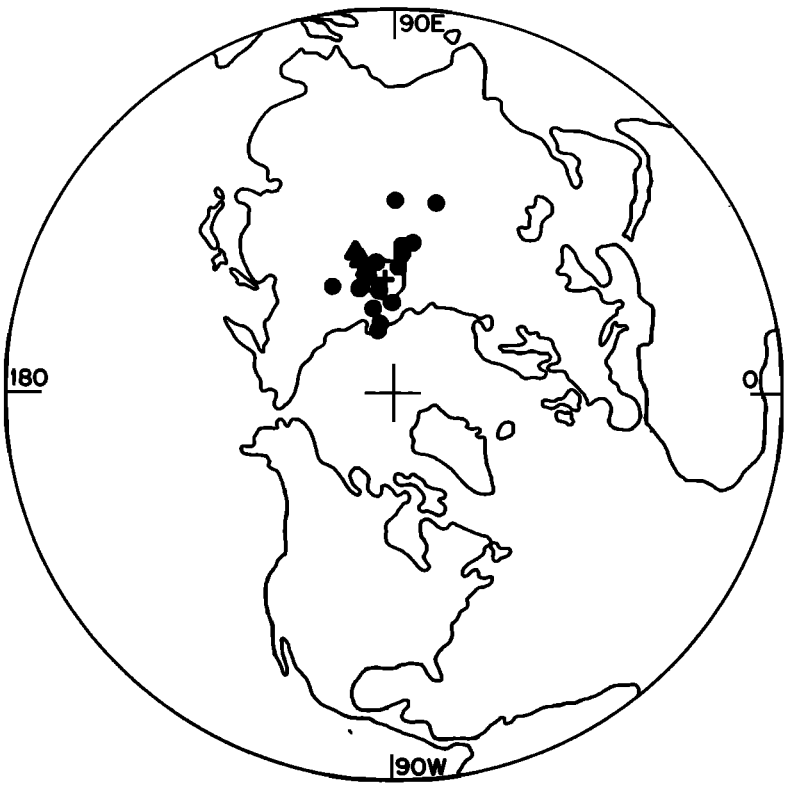


Fig. 4. Same poles as those shown in Figure 3 plotted on standard northern hemisphere equal-area projection. Triangles show southeastern Pennsylvania diabase poles. Plus symbol is mean pole for Upper Triassic igneous rocks from North America shown with its circle of confidence.

can be constructed. Results relevant to this study are reported by the various original authors in terms of samples, small sites, large sites, and entire small intrusions or groups of flows. Paleomagnetic poles are calculated for sites, single intrusions, groups of intrusions, groups of contiguous lava flows, and entire states. Clearly it is meaningless to compare a pole for a mean of five closely spaced samples from a single small intrusion with a pole representing the mean of all measurements from the state of Massachusetts, alloting both poles equal weight. In constructing Figures 3 and 4, I have arbitrarily selected pole calculations based on magnetization times likely to have been of the order of 10^4 years. A period of 10^4 years is useful in this respect, because it is long enough to be free of most of the effects of geomagnetic secular variation, but too short to obscure the effects of polar wandering. Figures 3 and 4 thus include poles for single large intrusions, contiguous groups of small intrusions, and groups of several lava flows. Some of these poles are based on too few sites to be greatly reliable, and others are suspected by the original investigators to represent too little time to have completely averaged out the secular variation. Both these causes may have introduced random scatter not related to motion of the axial dipole during magnetization of the eastern Upper Triassic igneous rocks. The dispersion of points thus may be too large, but the mean probably is unaffected. The mean of poles for Upper Triassic igneous rocks lies at 66°N, 95.5°E, with $N = 18$, $R = 17.77$, and an angular standard deviation of 9.2°.

The distribution of paleomagnetic poles shown in Figure 3 is markedly elongate, suggesting that the pole drifted while the Upper Triassic rocks were becoming magnetized. This

fact was noted previously by *de Boer* [1968], who used differences in inclination, reflecting differences in paleolatitude, to distinguish episodes of igneous activity in the Late Triassic record of the Central Appalachians. Figure 3 suggests that a plot of paleomagnetic poles might also be suitable for this purpose. In using Figure 3 for correlation, it is well to remember that uncertainties in individual pole positions (as measured by the angle α_{95}) range up to 10° or more.

Figure 3 suggests that apparent polar wandering during the period represented by Upper Triassic igneous rocks in eastern North America amounted to several tens of degrees. Figure 4 of *Opdyke and Wensink* [1966] can be interpreted similarly as evidence of substantial polar wandering relative to North America during the Jurassic. The general impression that the Mesozoic was a time of considerable relative motion between North America and the pole is substantiated by examination of any recent summary of North American paleomagnetic data [see, for instance, *Larochelle,* 1968, Figure 4]. By contrast, the southern continents show relatively little motion relative to the pole from the late Paleozoic to the Cretaceous. Paleomagnetic poles for Australia for the late Paleozoic and Mesozoic fall in a single group [*Irving,* 1966], and the same is true for Mesozoic poles for Africa [*McElhinney et al.,* 1968]. *Vilas and Valencio* [1970] indicate that the Triassic was a period of little or no motion of the pole relative to South America, and on the basis of available evidence it appears that the Jurassic might have been similarly static. In his summary of Gondwanaland paleomagnetism, *Briden* [1967] suggests that motion of the pole relative to the five southern continents was negligible for a period of about 10^8 years, beginning approximately in the Late Permian or Early Triassic and ending in the Late Cretaceous. This lack of evidence for marked polar shift relative to the southern continents, concurrent with the North American Mesozoic episode of apparent polar motion, suggests that there was little or no contribution from pure polar wandering (in the sense of relative motion between the earth's rotation axis and its deep interior) to the North American Mesozoic polar wandering curve. The latter conclusion was reached earlier by *Gromme' et al.* [1967] on the basis of their studies of Jurassic and Cretaceous paleomagnetism.

It has been suggested [*Irving,* 1966; see also *Hide,* 1967] that a correlation exists between high rates of apparent polar wandering and high frequencies of reversal of the geomagnetic field. On the basis of discussions by *Cox* [1969c] and *Goldreich and Toomre* [1969], as well as the original discussion by *Irving* [1966], it is possible to understand this correlation as a consequence of large-scale redistribution of mass, probably in the upper mantle. The trigger to this process might be a general convective overturn in the asthenosphere, accompanied by (or caused by) motion of lithospheric plates. Plate motion would in turn be accompanied by orogeny and would constitute pure continental drift, in the sense of motion of continental blocks relative to one another and to the pole, with no associated motion of the pole relative to the deep interior. However, large-scale redistribution of mass in the upper mantle would be likely to modify the instantaneous axis of figure of the earth, and polar wandering would ensue as the entire body of the earth adjusted to the new equilibrium position. According to the analysis of *Goldreich and Toomre* [1969], we might expect the rate of polar wandering to equal or exceed the rate of drift that caused it, and to be larger the larger the number of cells involved in the redistribution of mass. Thus active polar wandering and active continental drift may be causally linked. The link between apparent polar wandering (drift plus wandering) and frequency of geomagnetic reversal thus can be explained by the effect of polar wandering on magnetohydrodynamic processes in the outer core. Possibly interaction across the mantle-core boundary during rapid whole-body adjustment of the earth might cause an increase in the strength of nondipole sources of the geomagnetic field. Finally, as *Cox* [1968, 1969c] proposed, the resultant increase in the ratio of intensity of nondipole to dipole field would bring about an increase in the frequency of geomagnetic reversals.

The model outlined above predicts a correlation between secular variation, reversal frequency, worldwide rate of apparent polar wandering, and orogenesis. The model appears to

fit the brief period during the Late Triassic that saw the emplacement of diabase intrusions in the Gettysburg basin of Pennsylvania, and it is compatible with conditions during the Tertiary. It is hoped that the question of whether or not it has any general validity will be answered in the near future by detailed paleomagnetic studies, conducted on a world-wide basis, of geological units that are very precisely dated.

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